HeartLens | Phase 1 ML evaluation

Binary heart disease classification across four UCI hospitals

CSAI Project Team

Educational model • Source data: UCI Heart Disease

Problem and data

Target: num > 0 means disease present

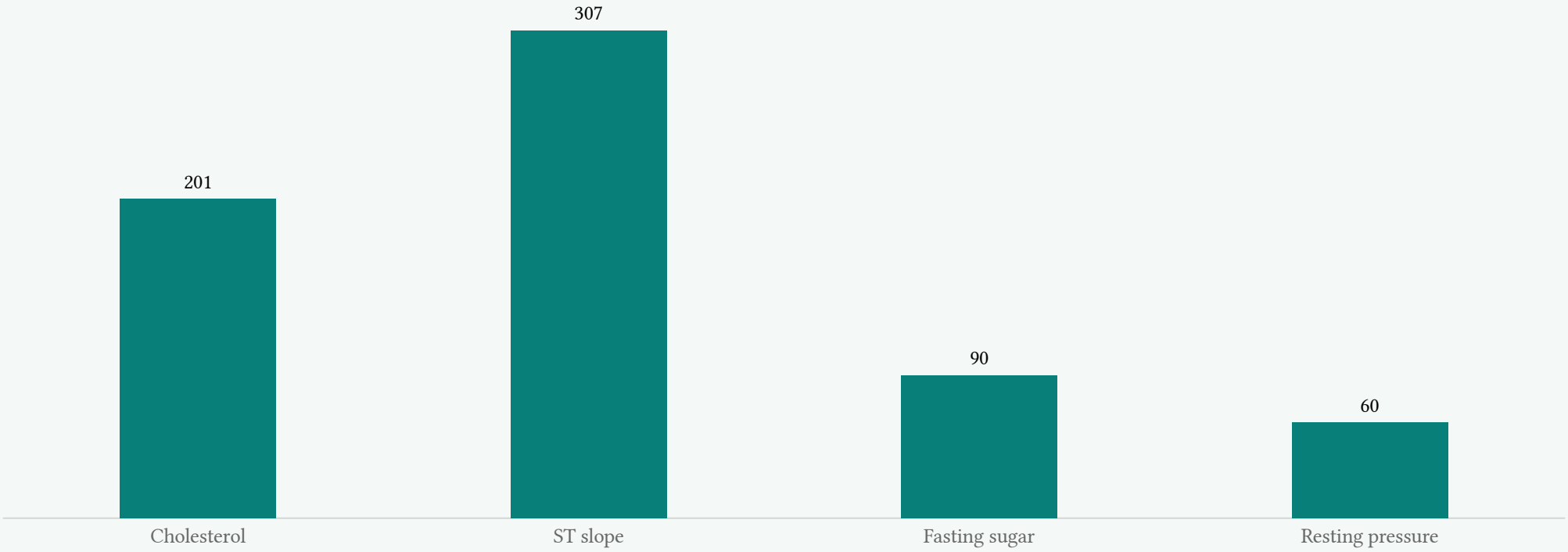
920 source rows → 918 after duplicate removal

508 positive and 410 negative records

11 predictors; hospital kept for generalization checks

Source: UCI Heart Disease, archive.ics.uci.edu/dataset/45/heart+disease

Data quality and missingness



Zeros in cholesterol and resting pressure are treated as missing.

Source: artifacts/eda.json, generated from the supplied CSV

Leakage-controlled experiment

Remove duplicate records before splitting

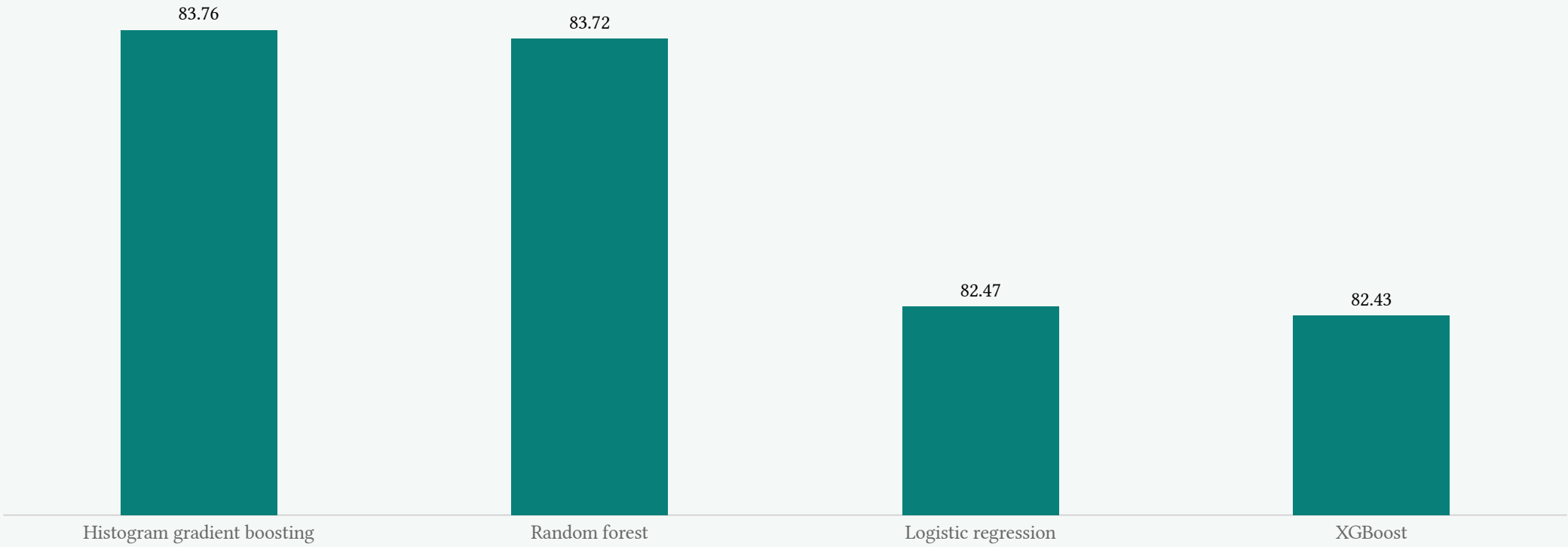
734 train / 184 test, stratified by binary label

Fit imputation, encoding and scaling inside each CV fold

Tune four model families with five-fold CV on train only

Choose cutoff from training predictions; test once

Cross-validation F1 (%)



Top two differ by 0.0004; fold variation is much larger.

Mean five-fold F1 on training records; selected model uses highest F1

Held-out test results

F1 0.836

Recall 0.873

ROC-AUC 0.901

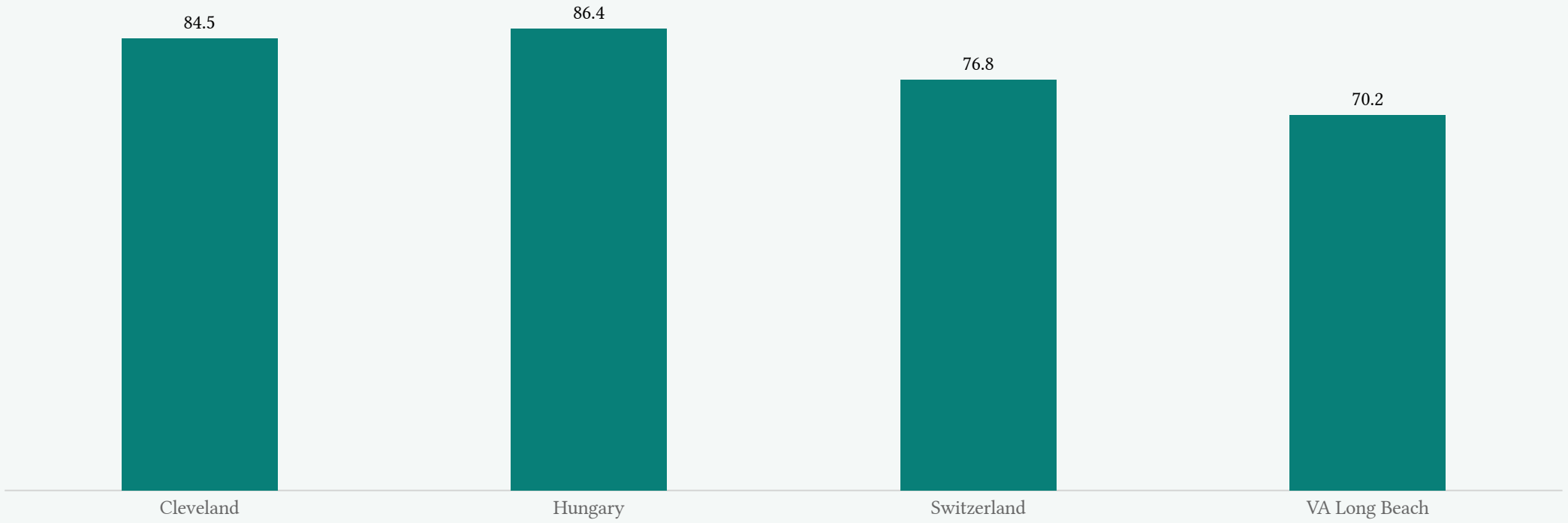
Cutoff 0.47

TN 60 • FP 22 • FN 13 • TP 89

False negatives remain; the score is not a diagnosis.

Source: artifacts/metrics.json; cutoff selected from training records only

ROC-AUC by hospital (%)



The four sites have different disease prevalence and missing-data patterns.

Separate leave-one-hospital-out experiments; held-out threshold = 0.50

Interpretation and limitations

Selected booster leads random forest by only 0.0004 F1

Chest-pain category has the largest test permutation AUC drop

Permutation importance is exploratory and affected by correlation

Small historical cohort; site shift limits transport

No prospective clinical or independent external validation

Reproduction and viva

python eda.py → dataset audit

python train.py → saved model and metrics

streamlit run app.py → prediction and evidence pages

Explain target, split, leakage control, metrics and hospital shift

Phase 1 contains classical ML only